- EC2 기반 -
MicroK8s Cluster 구축하기

무과금 Kubernetes 실습 환경 만들기 [Lv. 100]

- 깜지(김태지) -
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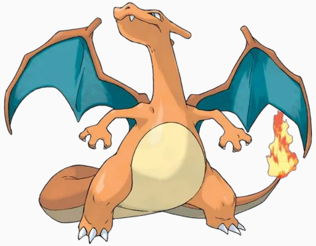
https://www.linkedin.com/in/taeji-kim-3b8a27268/ (링크드인)

발표자 소개

Level 1. 컴퓨터 정비기사

Level 2. 개발자

Level 3. SE/Cloud Engineer



목차

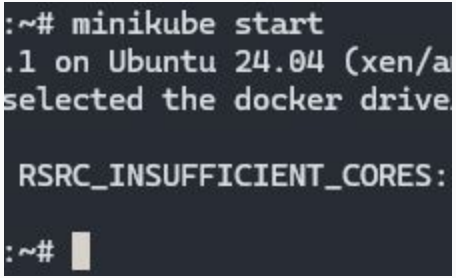
1. 쿠버네티스란?

컨테이너 오케스트레이션

- 고가용성
  - Self-Healing
  - Multi-Node Deployment
  - Load Balancing
  - ...
- 운영 효율성
  - HPA
  - Automated Deployment
  - Monitoring & Logging
  - ...

2. 우여곡절

Kubeadm → Minikube → K3s → MicroK8s



3. Why MicroK8s?

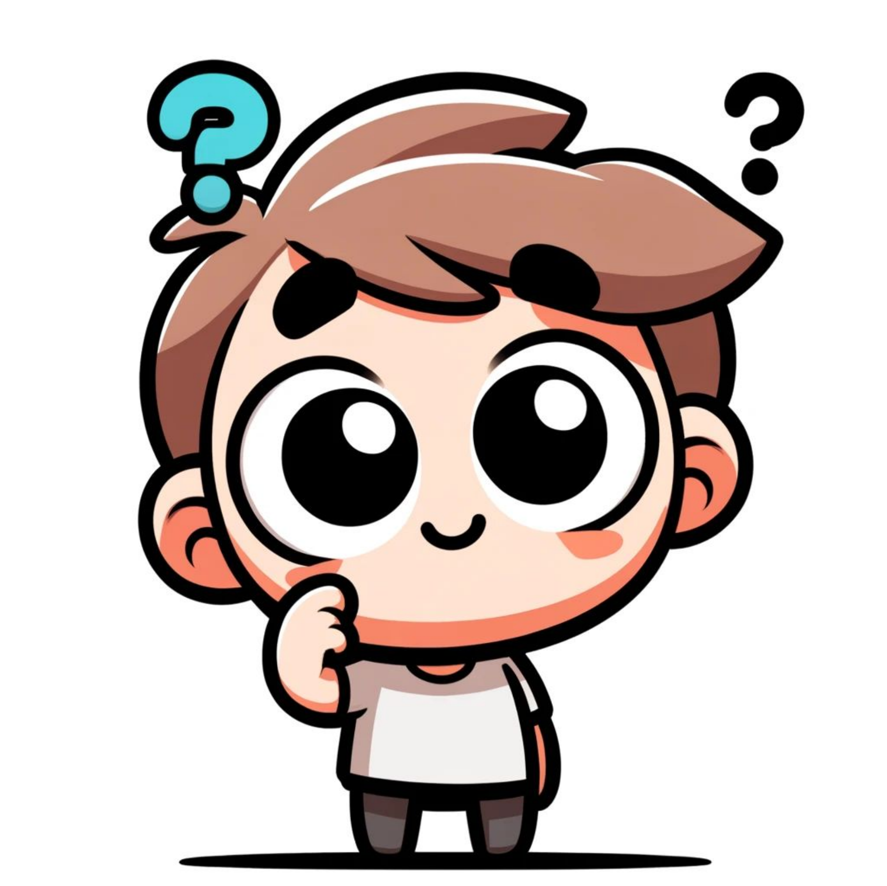
Easy to install

Add-on

Vanilla Kubernetes

Multi-node

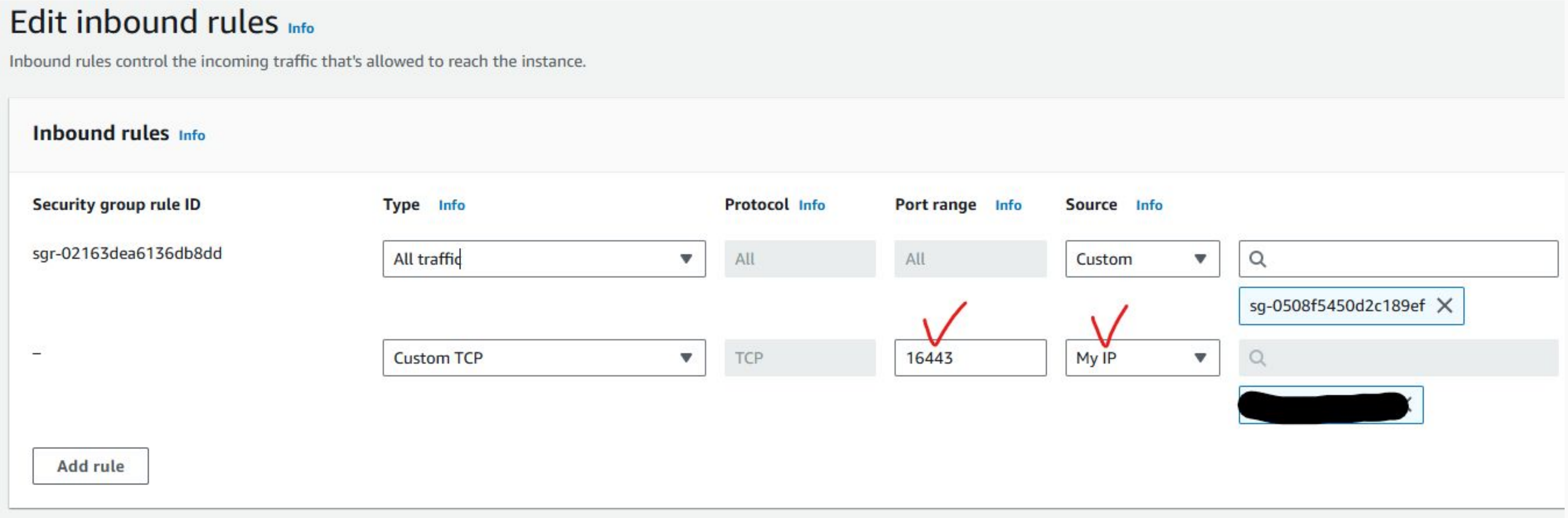
4. MicroK8s 구축 과정



— Local 에서 해당 클러스터에 명령하기 —

~~MicroK8s 구축 과정~~

4. Local 에서 해당 클러스터에 명령하기 – Edit inbound rules



kube-apiserver port: 16443

4. Local 에서 해당 클러스터에 명령하기 - cert-manager add-on 적용

cert-manager (add-on)

microk8s enable cert-manager

```
root@ip-10-0-0-139:~# microk8s status
microk8s is running
high-availability: no
  datastore master nodes: 127.0.0.1:19001
  datastore standby nodes: none
addons:
  enabled:
    cert-manager         # (core) Cloud native certi
    dns                  # (core) CoreDNS
    ha-cluster           # (core) Configure high ava
    helm                 # (core) Helm - the package
    helm3                # (core) Helm 3 - the packa
  disabled:
    cis-hardening        # (core) Apply CIS K8s hard
    community            # (core) The community addo
    dashboard            # (core) The Kubernetes das
    gpu                  # (core) Alias to nvidia ad
    host-access          # (core) Allow Pods connect
    hostpath-storage     # (core) Storage class; all
    ingress              # (core) Ingress controller
    kube-ovn             # (core) An advanced networ
    mayastor             # (core) OpenEBS MayaStor
    metallb              # (core) Loadbalancer for y
    metrics-server       # (core) K8s Metrics Server
```

4. Local 에서 해당 클러스터에 명령하기 - 인증서 재생성

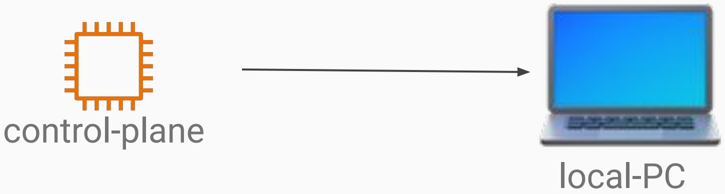
```
root@ip-10-0-0-139:/var/snap/microk8s/current/certs# cat csr.conf.template
[ req ]
default_bits = 2048
prompt = no
default_md = sha256
req_extensions = req_ext
distinguished_name = dn

[ dn ]
C = GB
ST = Canonical
L = Canonical
O = Canonical
OU = Canonical
CN = 127.0.0.1

[ req_ext ]
subjectAltName = @alt_names

[ alt_names ]
DNS.1 = kubernetes
DNS.2 = kubernetes.default
DNS.3 = kubernetes.default.svc
DNS.4 = kubernetes.default.svc.cluster
DNS.5 = kubernetes.default.svc.cluster.local
IP.1 = 127.0.0.1
IP.2 = 10.152.183.1
#MOREIPS

[ v3_ext ]
authorityKeyIdentifier=keyid,issuer:always
basicConstraints=CA:FALSE
keyUsage=keyEncipherment,dataEncipherment,digitalSignature
extendedKeyUsage=serverAuth,clientAuth
subjectAltName=@alt_names
```

```
root@ip-10-0-0-139:/var/snap/microk8s/current/certs# cat csr.conf.template
[ req ]
default_bits = 2048
prompt = no
default_md = sha256
req_extensions = req_ext
distinguished_name = dn

[ dn ]
C = GB
ST = Canonical
L = Canonical
O = Canonical
OU = Canonical
CN = 127.0.0.1

[ req_ext ]
subjectAltName = @alt_names

[ alt_names ]
DNS.1 = kubernetes
DNS.2 = kubernetes.default
DNS.3 = kubernetes.default.svc
DNS.4 = kubernetes.default.svc.cluster
DNS.5 = kubernetes.default.svc.cluster.local
IP.1 = 127.0.0.1
IP.2 = 10.152.183.1
IP.3 = 43.203.126.61
#MOREIPS

[ v3_ext ]
authorityKeyIdentifier=keyid,issuer:always
basicConstraints=CA:FALSE
keyUsage=keyEncipherment,dataEncipherment,digitalSignature
extendedKeyUsage=serverAuth,clientAuth
```

4. Local 에서 해당 클러스터에 명령하기 - kubeconfig 파일가져오기


```
root@ip-10-0-0-139:/var/snap/microk8s/current/credentials# cat client.config
apiVersion: v1
clusters:
- cluster:
    certificate-authority-data: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUREek
ERUSTBNRFV6TVRBd05UazBNMW9YRFRNMApNRFV5T1RBd05UazBNMW93RnpFVk1CTUdBMVVFQXd3TU1
5RTzNxRDhqVW1YcWpDNGRBMmx0dGh6Y254ejRBMkdieG93ejN0UFBVMnFJaEZoTGhCT2NtbTRHeDJk
5a0ZXQmNuQVVwdHQzOEhLZGpoRWZ2ZzFLRmUvZEVNaCtISjEKVDgvemFMMExRVmFxemJqMGdPZFZ6N
Qm8xTXdVVEFkCkJnTlZIUTRFRmdRVUp4SkZuSTgyZHhpRFYvaDc2WGNDWlloa2lUY3dId1lEVlIwak
llaamhqbGhTSER3L1dyN2xUNXN4c0NmTlN6Q1M4aU1jOHc1RU5ZdzdCSzdrUjJ5Cjg2cE5IeXM2OGl
d6TTlZVzE4OHp2K1FOTGJ3bGlGUgpwR3ZGM2JML1gwbFdnUXZUdWtmS1NFS3F4WHpMK2R0OFh1SUIy
4N1I4Qlo3UDNDVjIzT2VTKy9mbXc9PQotLS0tLUVORCBDRVJUSUZJQ0FURS0tLS0tCg==
    server: https://127.0.0.1:16443
  name: microk8s-cluster
contexts:
- context:
    cluster: microk8s-cluster
    user: admin
  name: microk8s
current-context: microk8s
kind: Config
preferences: {}
users:
- name: admin
  user:
    client-certificate-data: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUN6RENDQ
STBNRFV6TVRBd05UazBNMW9YRFRNMApNRFV5T1RBd05UazBNMW93S1RFT01Bd0dBMVVFQXd3RllXUn
UJRelBEZHJtdWVmM1h5blRKaU5XNFU3TEZkemdtajZGdEg2ZXlaemc0emxkRWxIY1V5djh0bG15bXA
oxdkhsQ2p4VHVSc01Td0xIVDRod2cwQ3BHQm9KVGppL2QKNjRSQmZ4cTM2Q2JyaUdZd1FzbDZOS21M
zYVdDY3lpCk1NYUdWSGNhMFFJREFRQUJNQTBHQ1NxR1NJYjNEUUVCQ3dVQUE0SUJBUUMvVWkwN2xvZ
cXYyTlM4dFc2Rm5YNitjMXdhekpFbDlIU1FkL0xncXQyaGhZeFVoeEFtSENICkxiYzduYVVhekJZen
jVoNjdXV3dpeHlsVVpjN2kweQo1VjRYN3g0RmF5WkNIVXc4aEFPT016aXdiWTVUUjlBRExGaXc1YkZ
    client-key-data: LS0tLS1CRUdJTiBSU0EgUFJJVkFURSBLRVktLS0tLQpNSUlFb3dJQkFBS
NHBNZEQ5YWdYU0lXVTZaYjV6SHAKZXpPZUZuSG9aMTNxSWNPVVFrc0tlTHpiZGpLc0NmR3lLcXVhL2
E16VFF5T3YzRDI2SUYyTDFBSTR6QWI0ckxHMER4QW8vYjBxa3dndEZEbXQ1NzVqWHFHWTE2ejc2NjY
Z5RmxhZU5hTGE1TGE1V2hzaGZRNjZxbmdDMnh1cHZyCmhZaUVrYkFZSTFBMDBleXBxU013ZjdrU25G
4WUQ3bQovYlh4aUJiWnFuWHgyV085Umt4QTZlWG5GcVpRZkdKeXF4OEcyK1JmRHUxaEVmZDJNdXF4K
NzJWWkhSMC9pVHp6TldFUHhpUnowQ1BkcWJSOU1mQy8ycDh0YlAxRVMrdgpVMEhhU1Q3a1FRcnEvME
WI0RlE5QmNDZ1lFQXpuYjMKYVlNbGdBZGdtQ29iM2VLU2xZM2JLc3Y2WDU0MUN3Q3lkbUNDR1BQNGo
pQMkI4RXVLWkFyaTVjTlUyZEJBcUR1V0E4NmNWY0NnWUVBdlMrVVFQMmRUYjF6blFjeS9xMDQKZEd3
wc2VIUlFTNG5BZ001ZWFyUTVUVUhGYjlRM3FrCjNqWlZDVmhpVTJJUUI3SnB4c0RVK244Q2dZQXhBc
bgpVL25uc3d2U0h3aDVzcDZaYkMvRFFpSVpwdnpZdjRTcGVXTWx4SEN4Z0I0bUJ4aHpLMUw2NXRkMT
0N5QWF4NFN1aTBxdm9rRGRHdEFoVE0ySDR0dUp4eDBqRGVsK29RQgppQUJDVndZY2lSRVhjWlAxUFB
```


/var/snap/microk8s/current/credentials/client.config



4. Local 에서 해당 클러스터에 명령하기 - kubeconfig 수정 후 확인

```
root@Zest ~/test# cat client.config
apiVersion: v1
clusters:
- cluster:
    certificate-authority-data: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS
ERUSTBNRFV6TVRBd05UazBNMW9YRFRNMApNRFV5T1RBd05UazBNMW93RnpFVk1CTUc
5RTzNxRDhqVW1YcWpDNGRBMmx0dGh6Y254ejRBMkdieG93ejN0UFBVMnFJaEZoTGh
5a0ZXQmNuQVVwdHQzOEhLZGpoRWZ2ZzFLRmUvZEVNaCtISjEKVDgvemFMMExRVmFx
Qm8xTXdVVEFkCkJnTlZIUTRFRmdRVUp4SkZuSTgyZHhpRFYvaDc2WGNDWlloa2lUY
llaamhqbGhTSER3L1dyN2xUNXN4c0NmTlN6Q1M4aU1jOHc1RU5ZdzdCSzdrUjJ5Cj
d6TTlZVzE4OHp2K1FOTGJ3bGlGUgpwR3ZGM2JML1gwbFdnUXZUdWtmS1NFS3F4WHp
4N1I4Qlo3UDNDVjIzT2VTKy9mbXc9PQotLS0tLUVORCBDRVJUSUZJQ0FURS0tLS0t
    server: https://43.203.126.61:16443
  name: microk8s-cluster
contexts:
- context:
    cluster: microk8s-cluster
    user: admin
  name: microk8s
current-context: microk8s
kind: Config
preferences: {}
users:
- name: admin
  user:
    client-certificate-data: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0t
STBNRFV6TVRBd05UazBNMW9YRFRNMApNRFV5T1RBd05UazBNMW93S1RFT01Bd0dBM
UJRelBEZHJtdWVmM1h5blRKaU5XNFU3TEZkemdtajZGdEg2ZXlaemc0emxkRWxIY1
oxdkhsQ2p4VHVSc01Td0xIVDRod2cwQ3BHQm9KVGppL2QKNjRSQmZ4cTM2Q2JyaUd
zYVdDY3lpCk1NYUdWSGNhMFFJREFRQUJNQTBHQ1NxR1NJYjNEUUVCQ3dVQUE0SUJB
cXYyTlM4dFc2Rm5YNitjMXdhekpFbDlIU1FkL0xncXQyaGhZeFVoeEFtSENICkxiY
jVoNjdXV3dpeHlsVVpjN2kweQo1VjRYN3g0RmF5WkNIVXc4aEFPT016aXdiWTVUUj
    client-key-data: LS0tLS1CRUdJTiBSU0EgUFJJVkFURSBLRVktLS0tLQpNS
NHBNZEQ5YWdYU0lXVTZaYjV6SHAKZXpPZUZuSG9aMTNxSWNPVVFrc0tlTHpiZGpLc
E16VFF5T3YzRDI2SUYyTDFBSTR6QWI0ckxHMER4QW8vYjBxa3dndEZEbXQ1NzVqWH
Z5RmxhZU5hTGE1TGE1V2hzaGZRNjZxbmdDMnh1cHZyCmhZaUVrYkFZSTFBMDBleXB
4WUQ3bQovYlh4aUJiWnFuWHgyV085Umt4QTZlWG5GcVpRZkdKeXF4OEcyK1JmRHUx
NzJWWkhSMC9pVHp6TldFUHhpUnowQ1BkcWJSOU1mQy8ycDh0YlAxRVMrdgpVMEhhU
```

```
root@Zest ~/test# kubectl delete pod/nginx-test --kubeconfig=./client.config
pod "nginx-test" deleted
root@Zest ~/test# kubectl get nodes --kubeconfig=./client.config
NAME            STATUS   ROLES    AGE   VERSION
ip-10-0-0-139   Ready    <none>   75m   v1.29.4
root@Zest ~/test# kubectl run nginx-test --image=nginx --kubeconfig=./client.config
pod/nginx-test created
root@Zest ~/test# kubectl get pods --kubeconfig=./client.config
NAME         READY   STATUS    RESTARTS   AGE
nginx-test   1/1     Running   0          18s
```

— 느낀점 —

— Q & A —

감사합니다!